Computer Vision Homework 1

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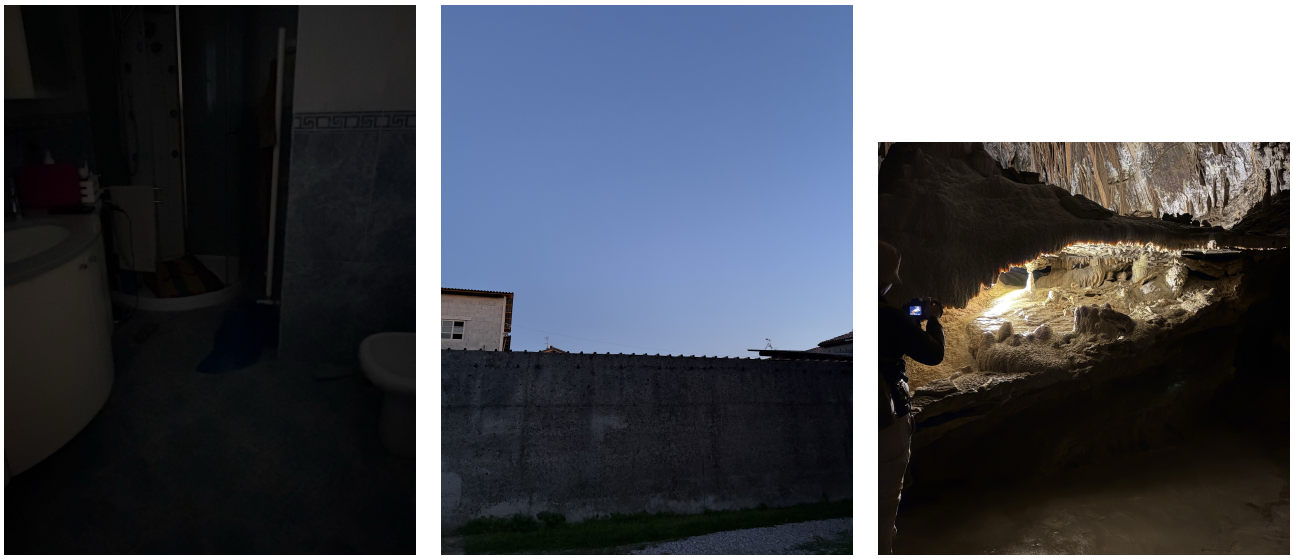
1 Introduction

This is a report for the first homework of Computer Vision 2026/2027.

2 Objective and Images used

The objective of this homework is the creation of a pipeline for image enhancement of visually degraded images (low-light, low-contrast, added noise).

The images used are the following:



(a) Low light (b) Outside (c) Cave

Figure 1: Original images used for the evaluation

Which have then been degraded by lowering the contrast and adding some gaussian noise, resulting in the following images:

(a) Low light (degraded) (b) Outside (degraded) (c) Cave (degraded)

Figure 2: Degraded images

The original images are only used for evaluation purposes, in real world scenarios noise-less, perfect references might not exist.

3 Methods

3.1 Linear Contrast Stretching

Linear contrast stretching is used in low-contrast images to spread the values over the whole available range (0-255). Each value is first normalized using the minimum and maximum values present in the image, and then multiplied by 255 to bring it back to the range 0-255.

$$L' = \frac{L - L_{\min}}{L_{\max} - L_{\min}} \cdot 255$$

For the purpose of this homework, this method has been applied to both each singular channel in the RGB color model, as well as on the luminance channel in the HLS color model.

3.2 Histogram Equalization

Histogram equalization is an image processing technique that takes the most frequent values and spreads them along all the possible values using a cumulative distribution. Ranges with many pixels get stretched apart, while ranges with few pixels get compressed, which usually improves contrast.

For this homework this method has been applied to the value channel in the HSV color model.

3.3 Bilateral filter + Gamma correction

Bilateral filtering is a smoothing filter that aims at reducing noise present in the image. In this homework it has been implemented along with gamma correction, which aims at making the

image generally brighter by moving up each pixel value's intensity using the gamma power law, meaning that darker pixels get much brighter than already bright pixels.

$$I' = 255 \cdot \left(\frac{I}{255}\right)^{\gamma}$$

Filtering is applied before gamma correction as noise can mess with the gamma correction step making it more visible in the darker regions.

4 Results

Method	low_light		outside		cave	
	PSNR (dB)	MSE	PSNR (dB)	MSE	PSNR (dB)	MSE
Degraded (baseline)	21.94	416.2	17.10	1269.3	18.11	1004.3
Linear stretch (RGB)	17.10	1268.2	17.05	1283.7	18.11	1004.3
Linear stretch (HLS)	13.28	3053.1	16.66	1403.5	18.15	996.7
Hist. equalization (HSV)	9.28	7682.6	17.22	1232.6	11.47	4633.3
Bilateral + gamma	30.36	59.8	19.69	697.7	20.66	558.9

Table 1: PSNR and MSE of each method with respect to the original image

4.1 Low Light

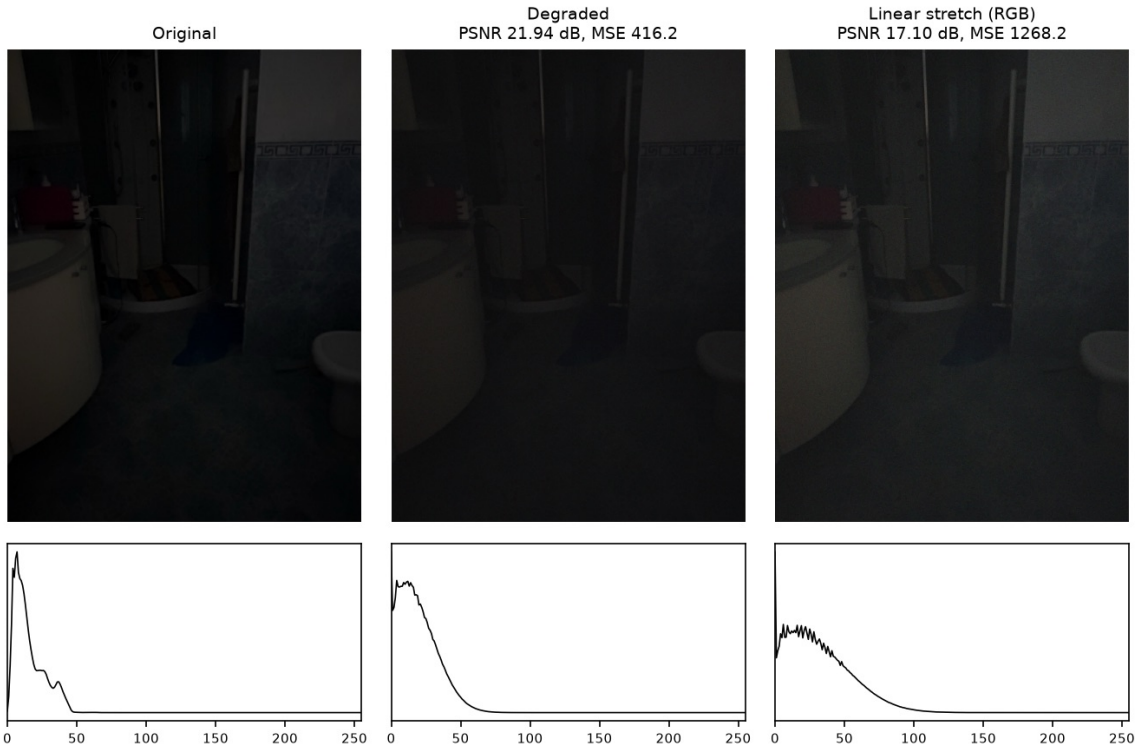


Figure 3: Low light: linear contrast stretching on each RGB channel

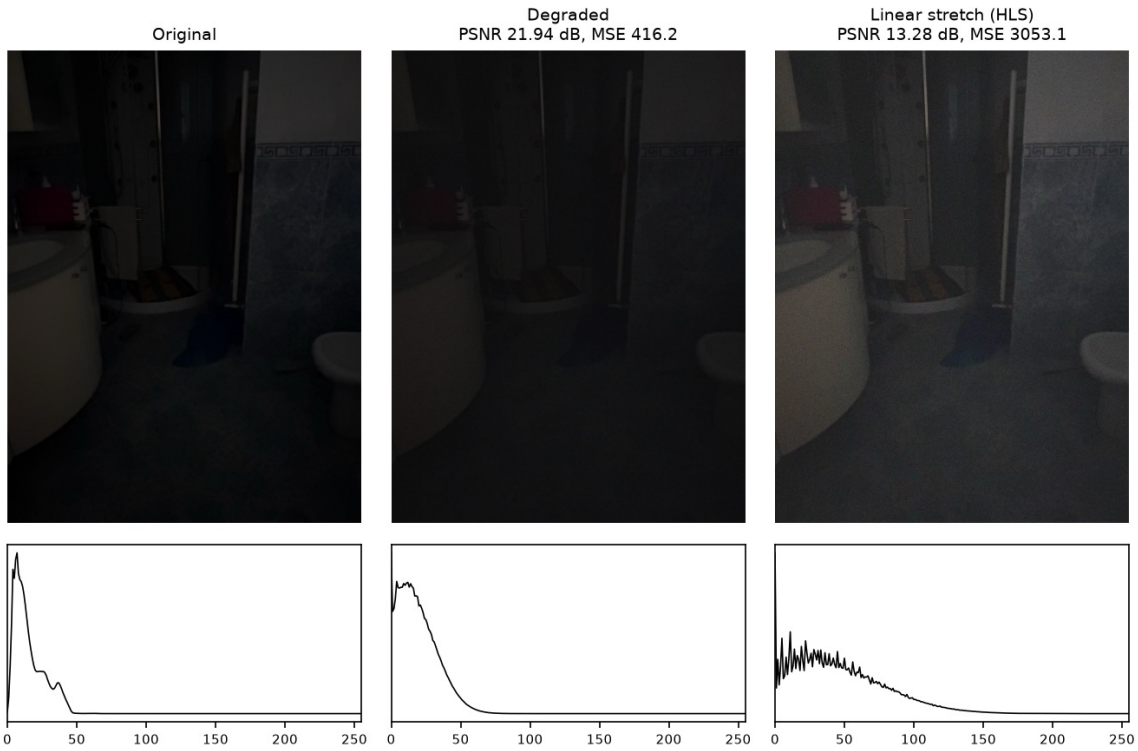


Figure 4: Low light: linear contrast stretching on the luminance channel (HLS)

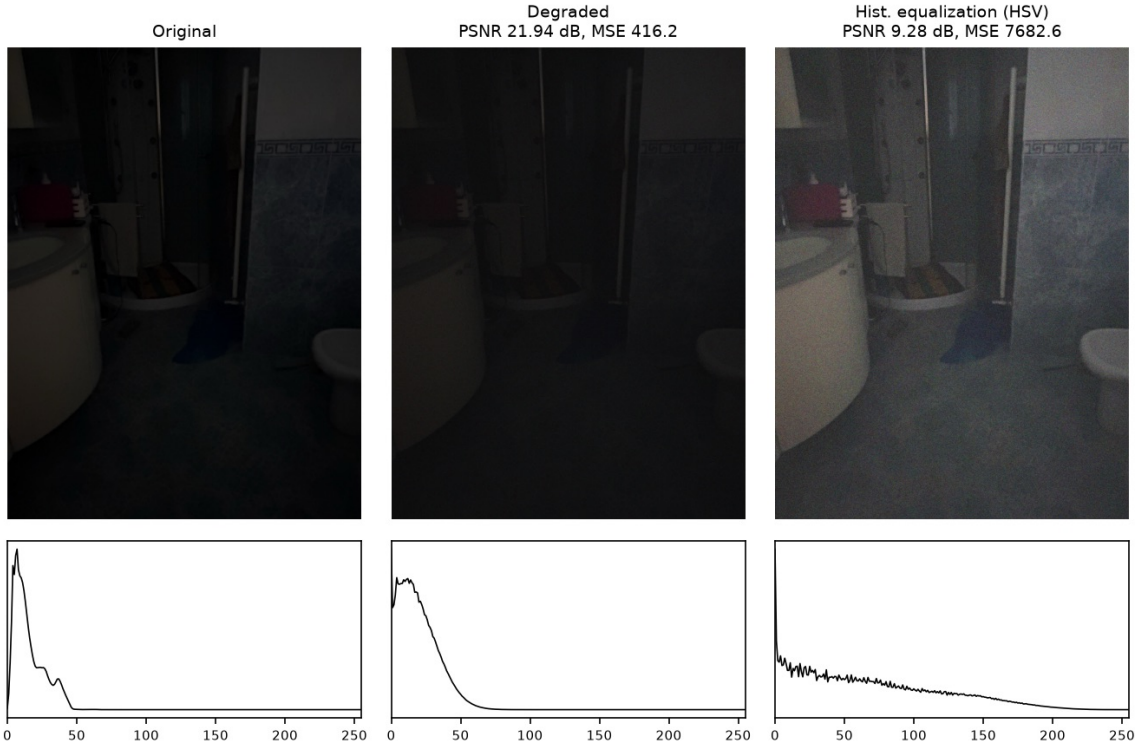


Figure 5: Low light: histogram equalization on the value channel (HSV)

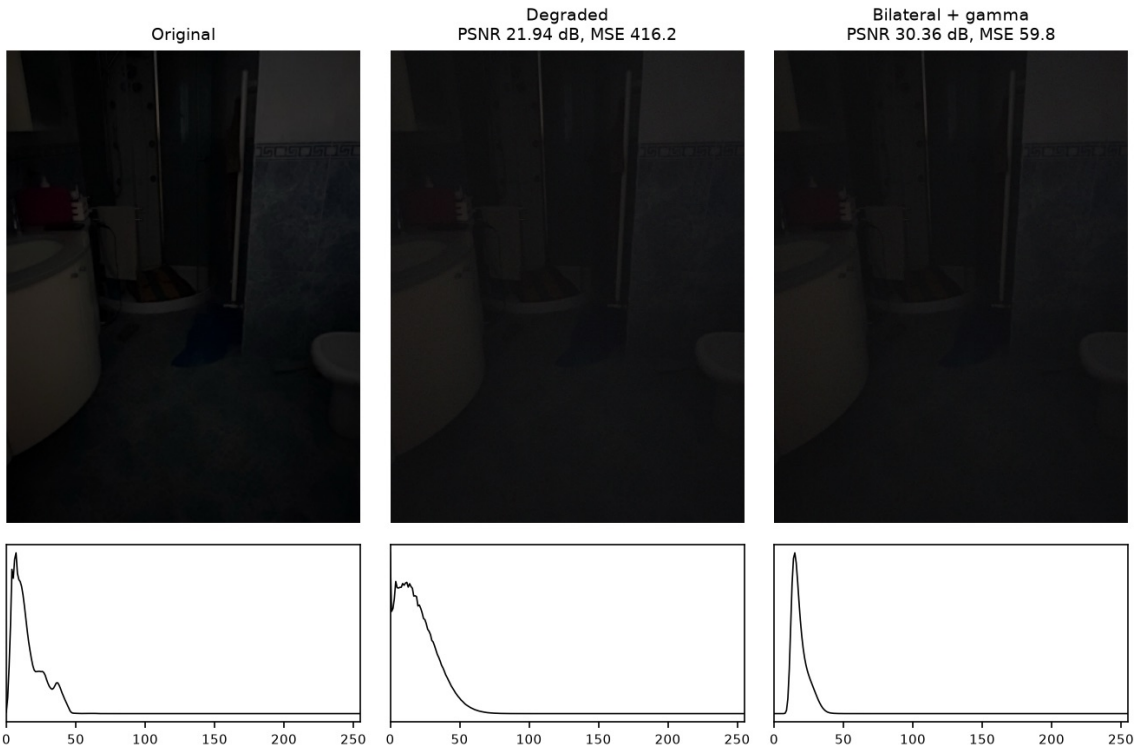


Figure 6: Low light: bilateral filter followed by gamma correction

4.2 Outside

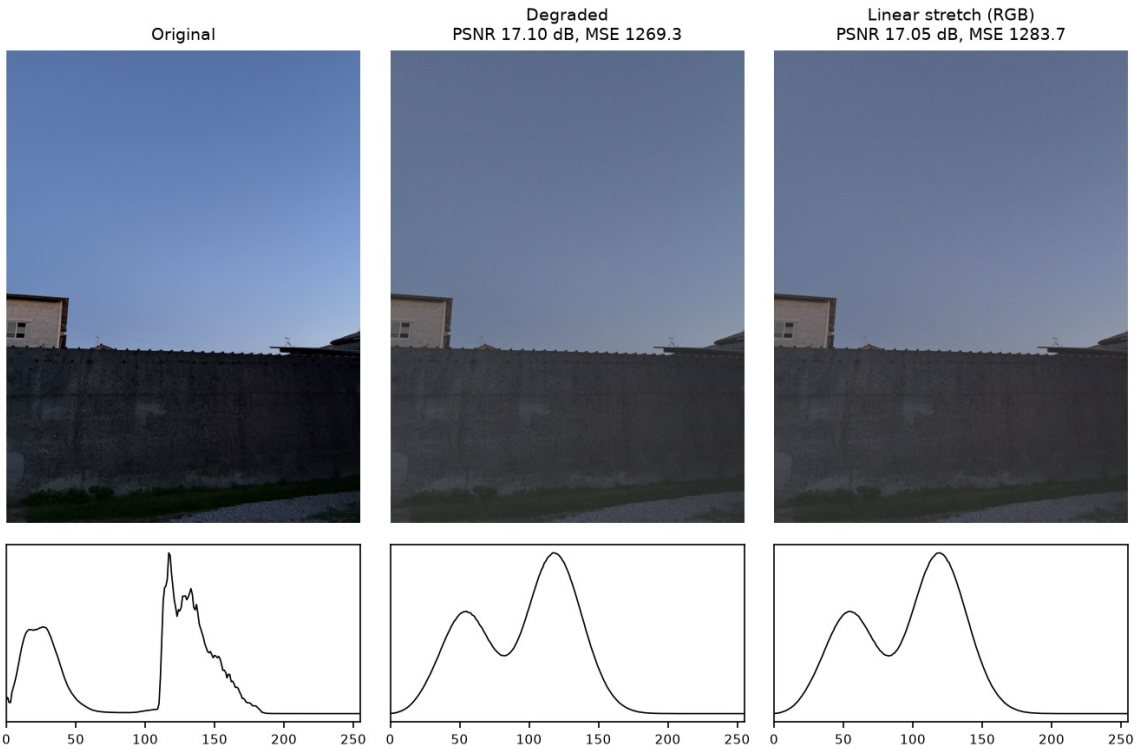


Figure 7: Outside: linear contrast stretching on each RGB channel

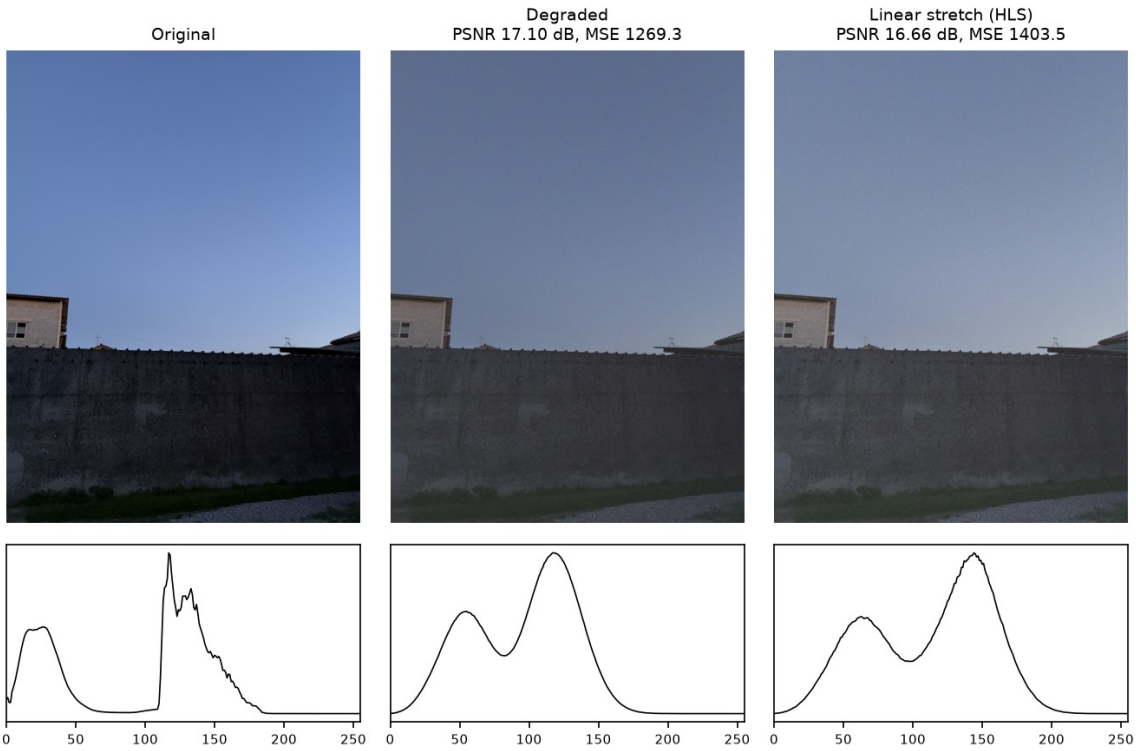


Figure 8: Outside: linear contrast stretching on the luminance channel (HLS)

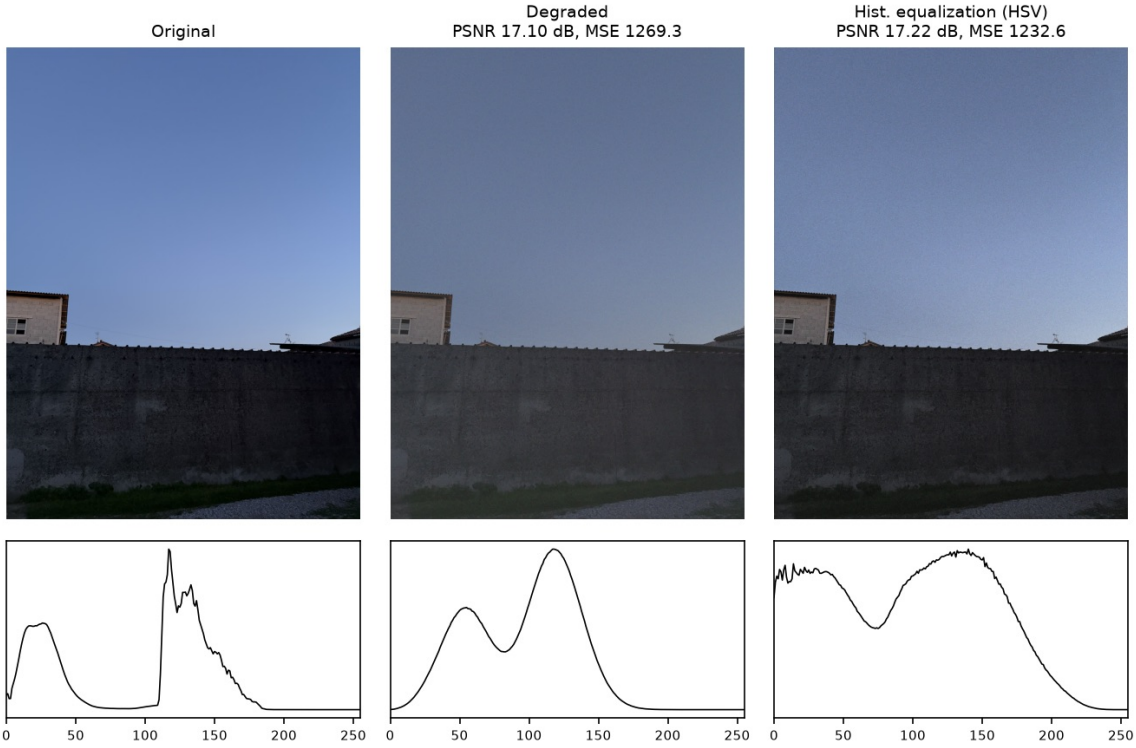


Figure 9: Outside: histogram equalization on the value channel (HSV)

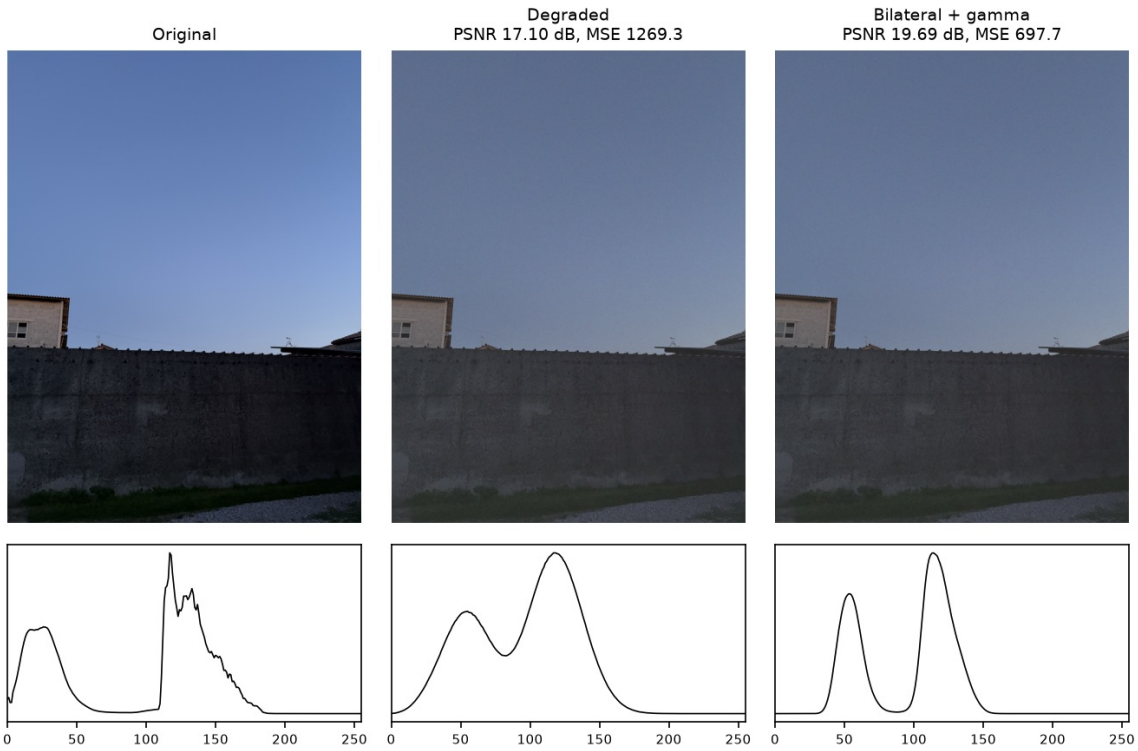


Figure 10: Outside: bilateral filter followed by gamma correction

4.3 Cave

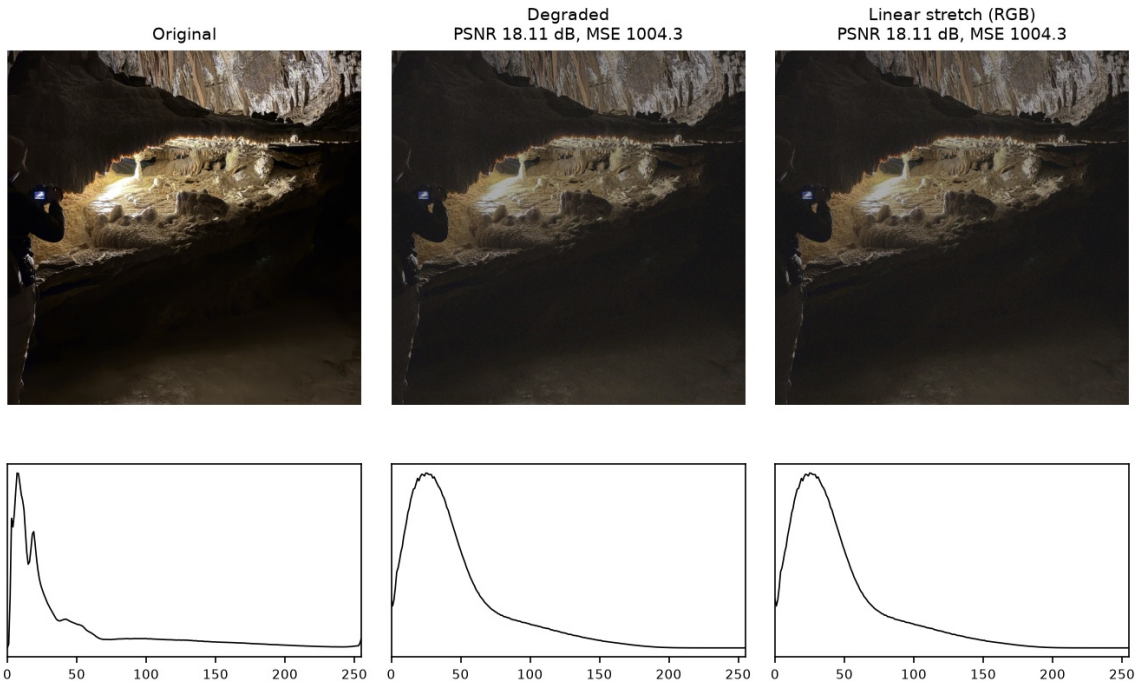


Figure 11: Cave: linear contrast stretching on each RGB channel

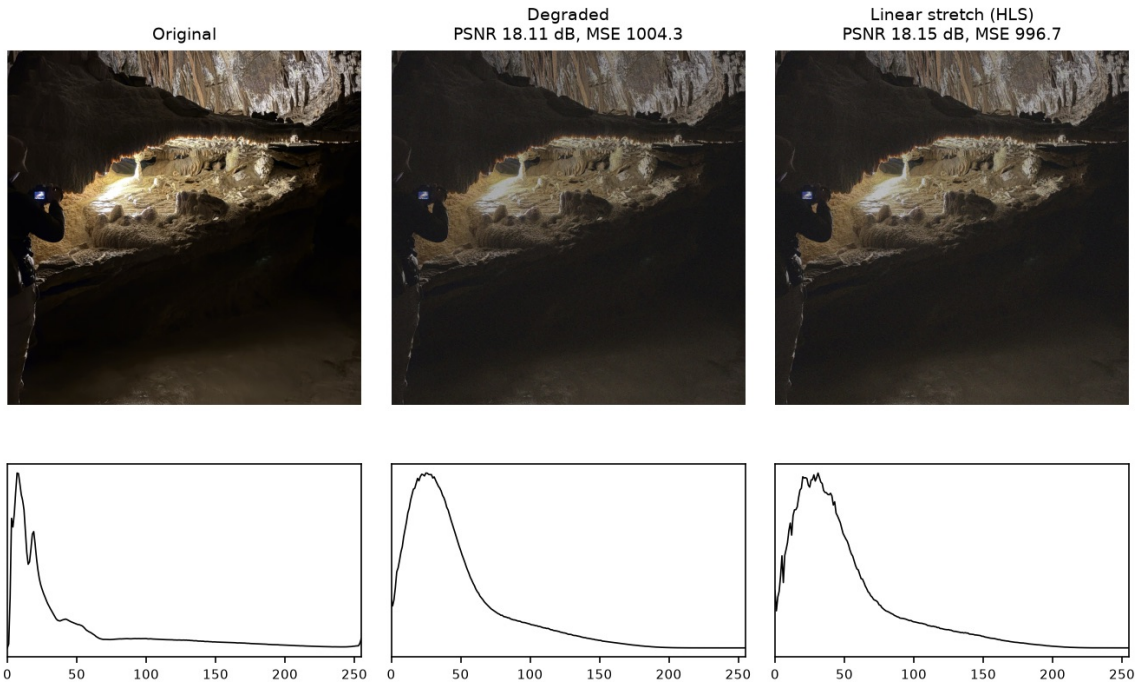


Figure 12: Cave: linear contrast stretching on the luminance channel (HLS)

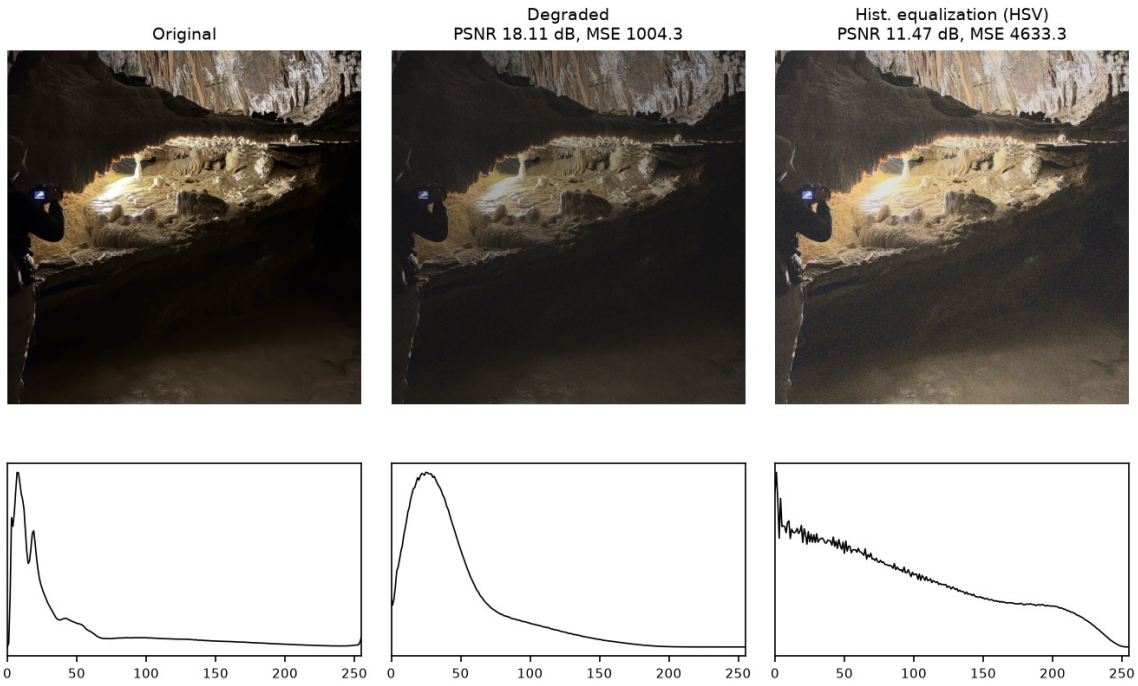


Figure 13: Cave: histogram equalization on the value channel (HSV)

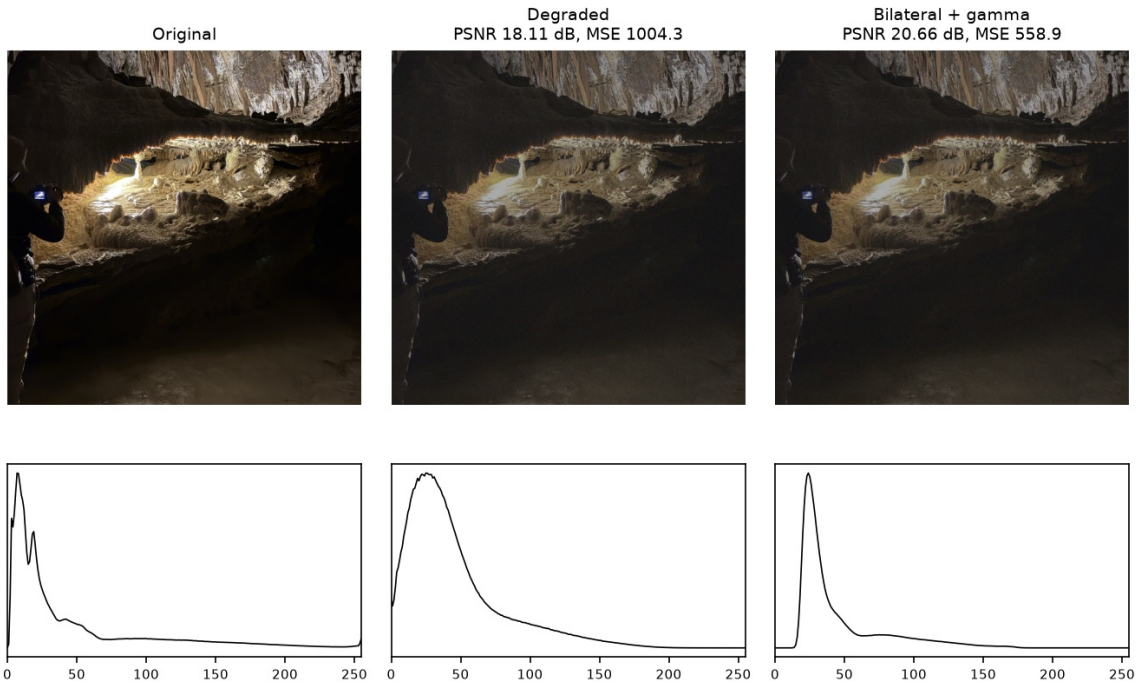


Figure 14: Cave: bilateral filter followed by gamma correction

5 Considerations

In all images we see that the better result in terms of PSNR and MSE is the bilateral filter + small gamma correction (gamma = 0.9). This is simply due to the fact that the added noise in the degraded image is responsible for most of the error. By removing it with smoothing we reduce its impact, however this might bring some side-effects such as blurred edges, in these specific cases and using bilateral filtering this effect is not as visible but it is present nonetheless.

The most visually pleasing results are given by the histogram equalization, which is able to bring out more details in the scene which are not really visible due to the low-light conditions. An especially pleasing result is the Cave example where the patterns in the rock at the top left are more visible. Another great case is in the low light image, where pieces of the scene are clearly visible unlike both the degraded and original images.

Both cases have the lowest PSNR. This is due to the fact that the original images are dark, while the enhanced images are lighter, PSNR is a pixel-by-pixel comparison with the original as such the error is big.

While still being visually pleasing the noise is much more visible in these enhanced images.

Between the two linear stretching methods little changes in terms of visual quality and error metrics. One especially interesting case is the cave image, where the RGB linear stretch results in the same exact PSNR and MSE as the degraded image. This means that the degraded image already uses the maximum possible range of values 0-255.

The low light image is an exception as the linear stretching applied in the HLS color model behaves more similarly to histogram equalization, being able to bring out details in the scene at the cost of PSNR and MSE.

The code can be found at: https://github.com/Zmuv/CV-26-27_homeworks